

Pantheon Report

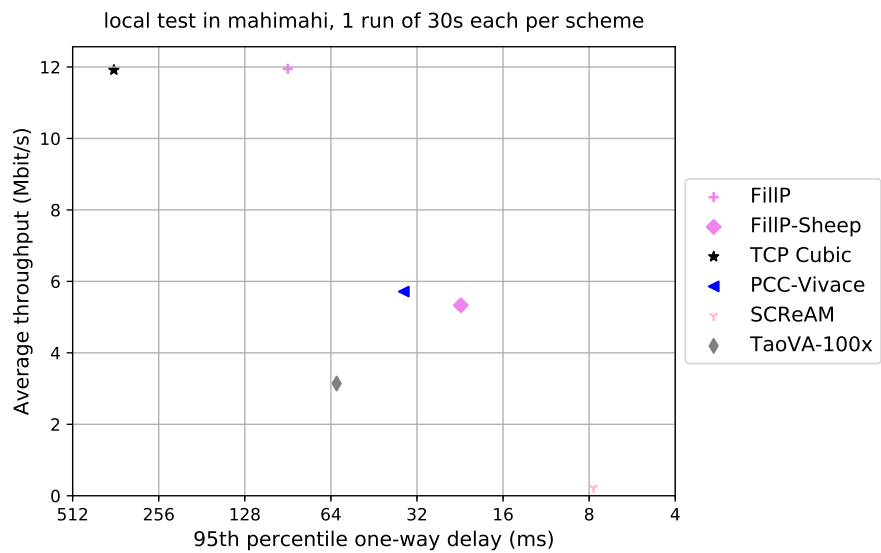
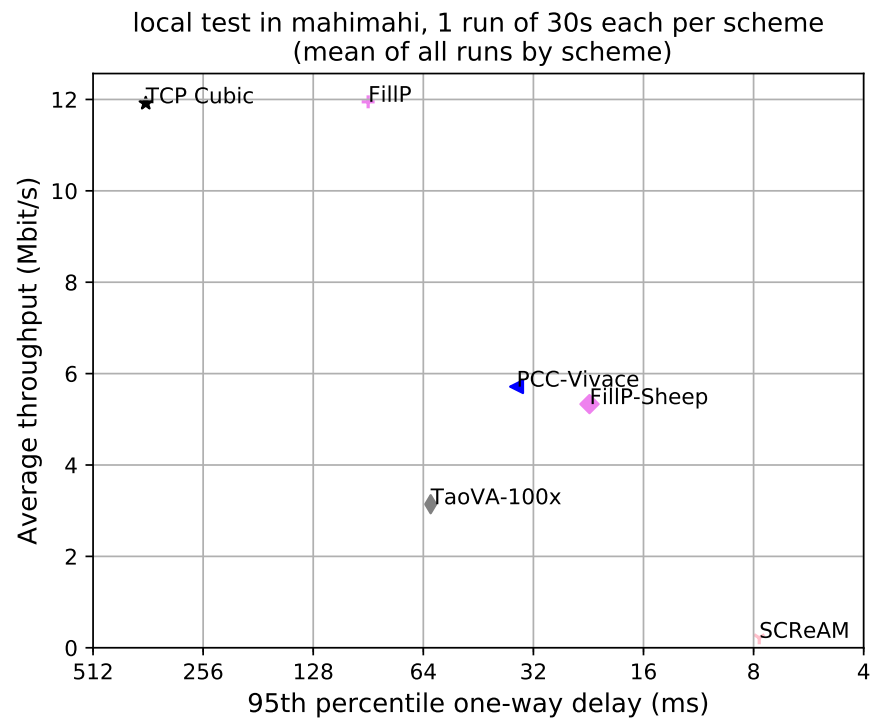
Generated at 2025-04-18 01:32:08 (UTC).
Tested in mahimahi: mm-link 12mbps.trace 12mbps.trace
Repeated the test of 17 congestion control schemes once.
Each test lasted for 30 seconds running 1 flow.

System info:

Linux 5.4.0-84-generic
net.core.default_qdisc = fq_codel
net.core.rmem_default = 212992
net.core.rmem_max = 212992
net.core.wmem_default = 212992
net.core.wmem_max = 212992
net.ipv4.tcp_rmem = 4096 131072 6291456
net.ipv4.tcp_wmem = 4096 16384 4194304

Git summary:

branch: master @ 23e738ce5acae1d36e321886cd613b0b9401ac11
third_party/fillp @ d6da1459332fcee56963885d7eba17e6a32d4519
third_party/fillp-sheep @ 0e5bb722943babcd2b090d2c64fcd45e12e923f9
third_party/genericCC @ d0153f8e594aa89e93b032143cedbdfef58e562f4
third_party/indigo @ 463d89b09699a57bfdfbae351646df6a60040b90
third_party/libutp @ b3465b942e2826f2b179eaab4a906ce6bb7cf3cf
third_party/pantheon-tunnel @ f866d3f58d27afd942717625ee3a354cc2e802bd
third_party/pcc @ 1afc958fa0d66d18b623c091a55fec872b4981e1
third_party/pcc-experimental @ cd43e34e3f5f5613e8acd08fab92c4eb24f974ab
third_party/proto-quic @ 77961f1a82733a86b42f1bc8143ebc978f3cff42
third_party/scream-reproduce @ f099118d1421aa3131bf11ff1964974e1da3bdb2
third_party/sprout @ 366e35c6178b01e31d4a46ad18c74f9415f19a26
third_party/verus @ d4b447ea74c6c60a261149af2629562939f9a494
third_party/vivace @ 2baf86211435ae071a32f96b7d8c504587f5d7f4
third_party/webrtc @ 3f0cc2a9061a41b6f9dde4735770d143a1fa2851



scheme	# runs	mean avg tput (Mbit/s) flow 1	mean 95th-%ile delay (ms) flow 1	mean loss rate (%) flow 1
TCP BBR	0	N/A	N/A	N/A
Copa	0	N/A	N/A	N/A
TCP Cubic	1	11.92	367.40	1.86
FillP	1	11.95	90.57	0.28
FillP-Sheep	1	5.33	22.48	0.00
Indigo	0	N/A	N/A	N/A
LEDBAT	0	N/A	N/A	N/A
PCC-Allegro	0	N/A	N/A	N/A
PCC-Expr	0	N/A	N/A	N/A
QUIC Cubic	0	N/A	N/A	N/A
SCReAM	1	0.22	7.72	0.00
Sprout	0	N/A	N/A	N/A
TaoVA-100x	1	3.14	61.10	0.00
TCP Vegas	0	N/A	N/A	N/A
Verus	0	N/A	N/A	N/A
PCC-Vivace	1	5.71	35.58	0.01
WebRTC media	0	N/A	N/A	N/A

Run 1: Statistics of TCP BBR

Start at: 2025-04-18 01:25:45

End at: 2025-04-18 01:26:15

Run 1: Report of TCP BBR — Data Link

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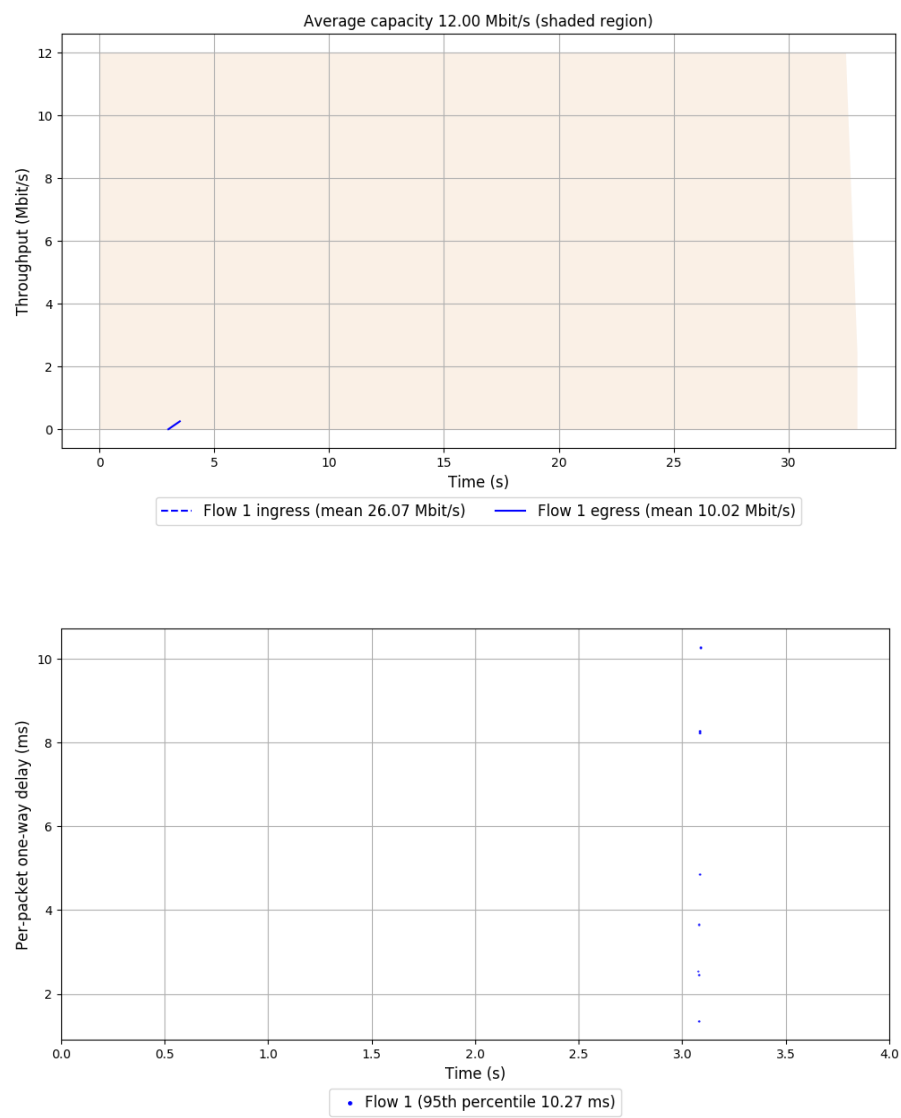
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Run 1: Statistics of Copa

Start at: 2025-04-18 01:21:52

End at: 2025-04-18 01:22:22

Run 1: Report of Copa — Data Link



Run 1: Statistics of TCP Cubic

Start at: 2025-04-18 01:23:32

End at: 2025-04-18 01:24:02

Below is generated by plot.py at 2025-04-18 01:32:05

Datalink statistics

-- Total of 1 flow:

Average capacity: 12.00 Mbit/s

Average throughput: 11.92 Mbit/s (99.3% utilization)

95th percentile per-packet one-way delay: 367.396 ms

Loss rate: 1.86%

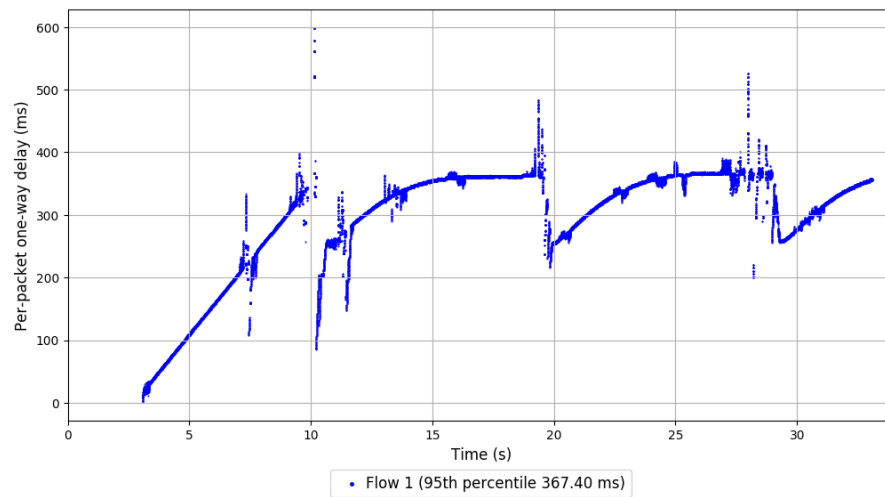
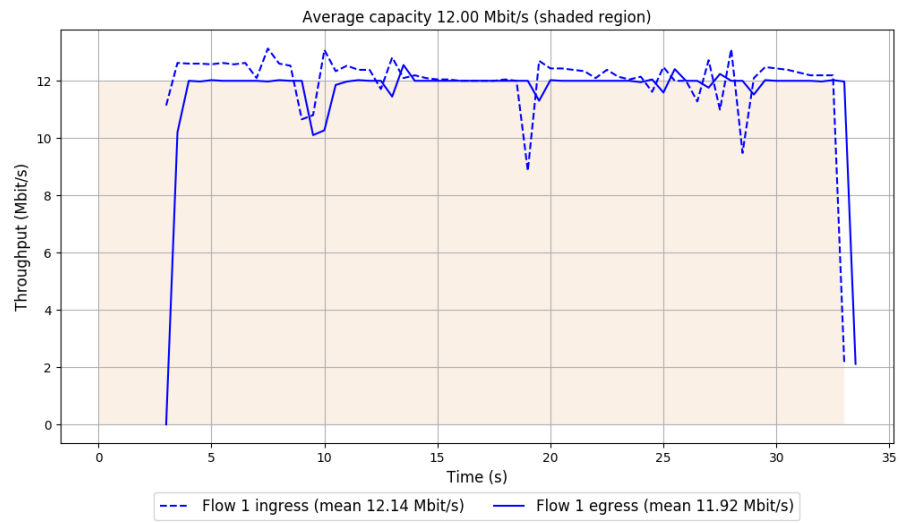
-- Flow 1:

Average throughput: 11.92 Mbit/s

95th percentile per-packet one-way delay: 367.396 ms

Loss rate: 1.86%

Run 1: Report of TCP Cubic — Data Link

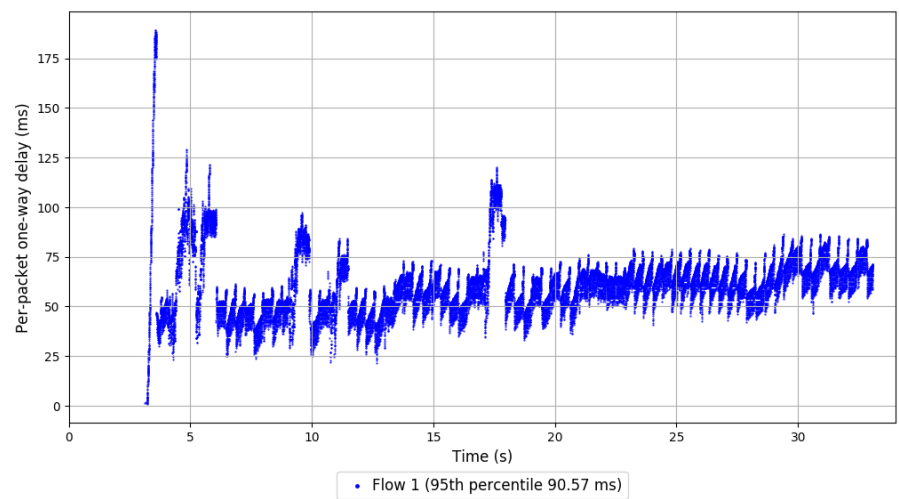
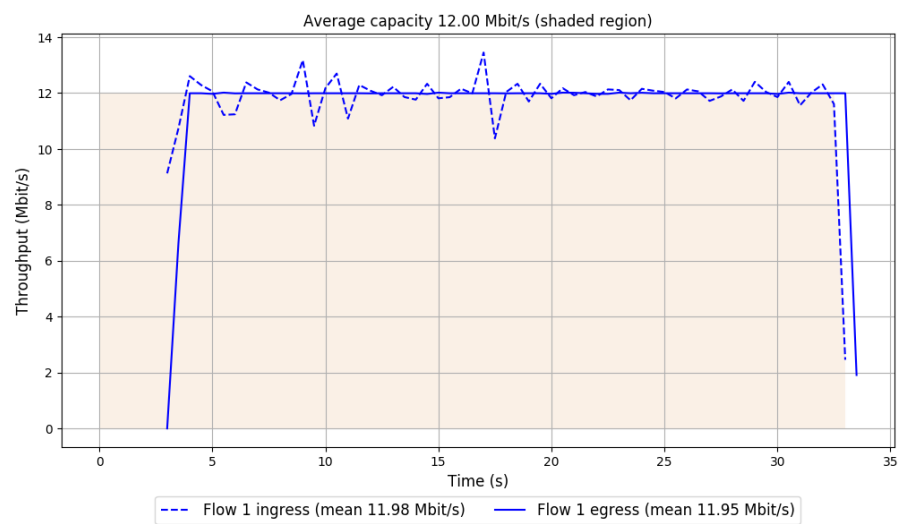


```
Run 1: Statistics of FillP

Start at: 2025-04-18 01:26:52
End at: 2025-04-18 01:27:22

# Below is generated by plot.py at 2025-04-18 01:32:06
# Datalink statistics
-- Total of 1 flow:
Average capacity: 12.00 Mbit/s
Average throughput: 11.95 Mbit/s (99.6% utilization)
95th percentile per-packet one-way delay: 90.574 ms
Loss rate: 0.28%
-- Flow 1:
Average throughput: 11.95 Mbit/s
95th percentile per-packet one-way delay: 90.574 ms
Loss rate: 0.28%
```

Run 1: Report of FillP — Data Link

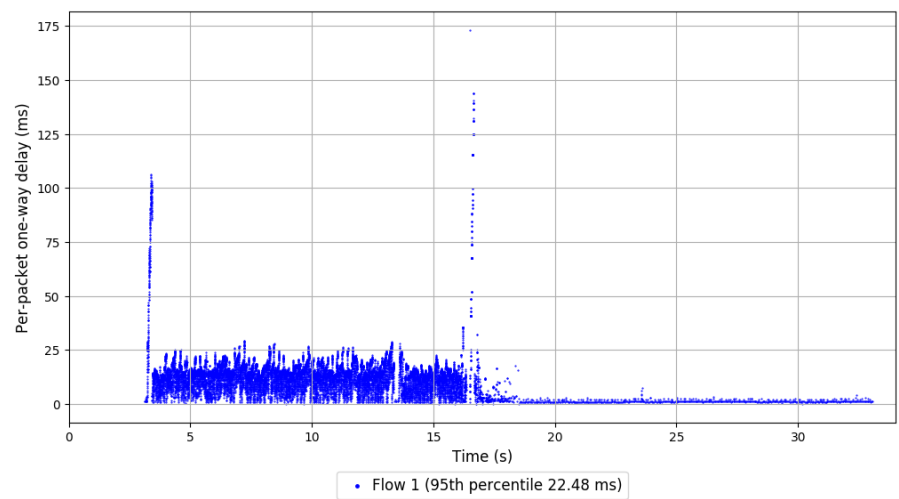
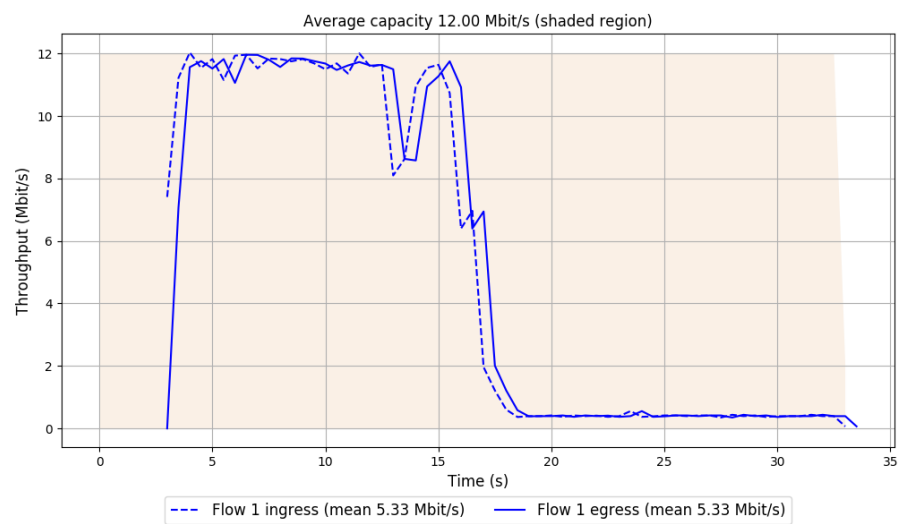


```
Run 1: Statistics of FillP-Sheep

Start at: 2025-04-18 01:21:18
End at: 2025-04-18 01:21:48

# Below is generated by plot.py at 2025-04-18 01:32:06
# Datalink statistics
-- Total of 1 flow:
Average capacity: 12.00 Mbit/s
Average throughput: 5.33 Mbit/s (44.4% utilization)
95th percentile per-packet one-way delay: 22.478 ms
Loss rate: 0.00%
-- Flow 1:
Average throughput: 5.33 Mbit/s
95th percentile per-packet one-way delay: 22.478 ms
Loss rate: 0.00%
```

Run 1: Report of FillP-Sheep — Data Link



Run 1: Statistics of Indigo

Start at: 2025-04-18 01:22:25

End at: 2025-04-18 01:22:55

Run 1: Report of Indigo — Data Link

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Run 1: Statistics of LEDBAT

Start at: 2025-04-18 01:29:39

End at: 2025-04-18 01:30:09

Run 1: Report of LEDBAT — Data Link

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Run 1: Statistics of PCC-Allegro

Start at: 2025-04-18 01:28:32

End at: 2025-04-18 01:29:02

Run 1: Report of PCC-Allegro — Data Link

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Figure is missing

Run 1: Statistics of PCC-Expr

Start at: 2025-04-18 01:30:12

End at: 2025-04-18 01:30:42

Run 1: Report of PCC-Expr — Data Link

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Figure is missing

Run 1: Statistics of QUIC Cubic

Start at: 2025-04-18 01:27:59

End at: 2025-04-18 01:28:29

Run 1: Report of QUIC Cubic — Data Link

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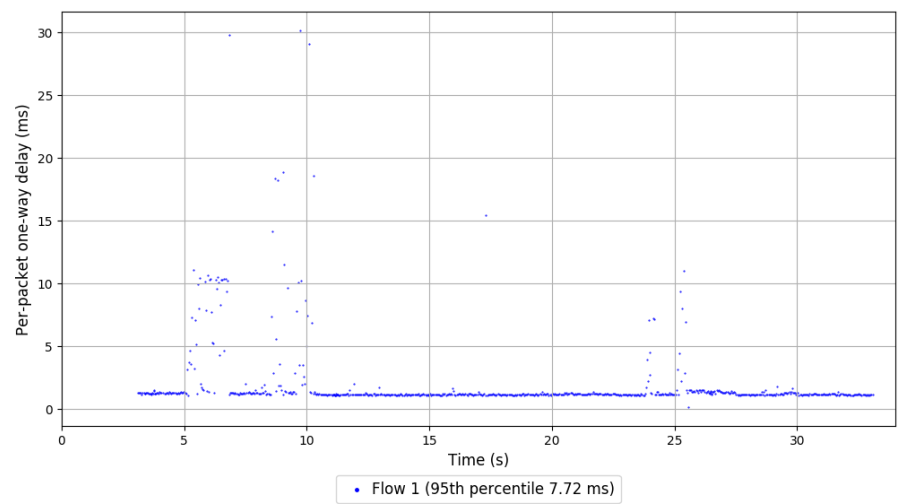
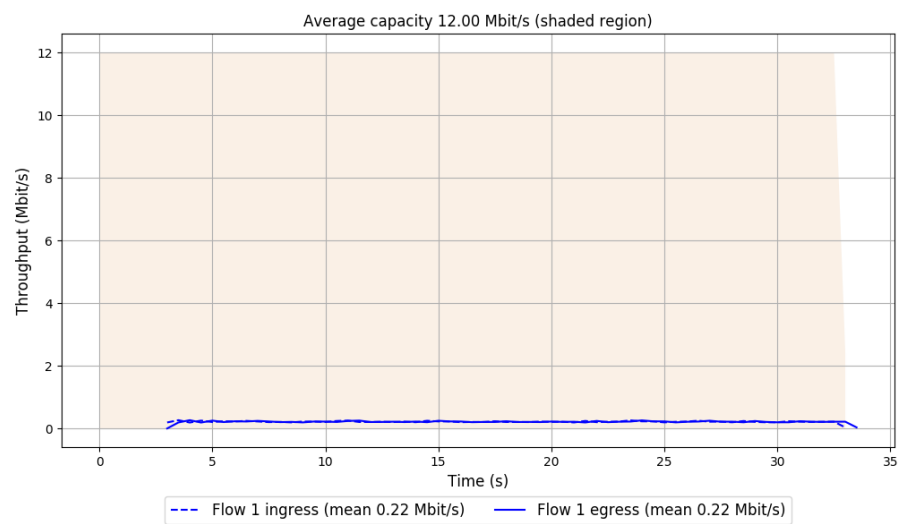
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```
Run 1: Statistics of SCReAM

Start at: 2025-04-18 01:27:26
End at: 2025-04-18 01:27:56

# Below is generated by plot.py at 2025-04-18 01:32:06
# Datalink statistics
-- Total of 1 flow:
Average capacity: 12.00 Mbit/s
Average throughput: 0.22 Mbit/s (1.8% utilization)
95th percentile per-packet one-way delay: 7.720 ms
Loss rate: 0.00%
-- Flow 1:
Average throughput: 0.22 Mbit/s
95th percentile per-packet one-way delay: 7.720 ms
Loss rate: 0.00%
```


Run 1: Report of SReAM — Data Link



Run 1: Statistics of Sprout

Start at: 2025-04-18 01:24:05

End at: 2025-04-18 01:24:35

Run 1: Report of Sprout — Data Link

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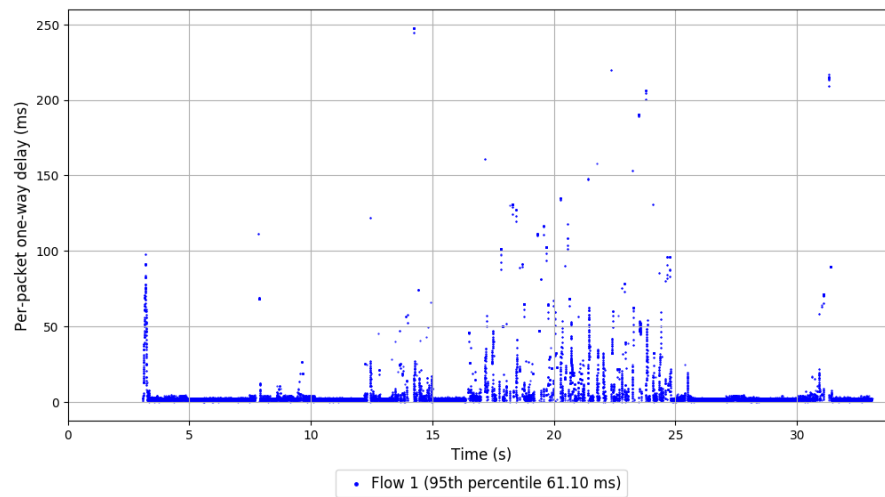
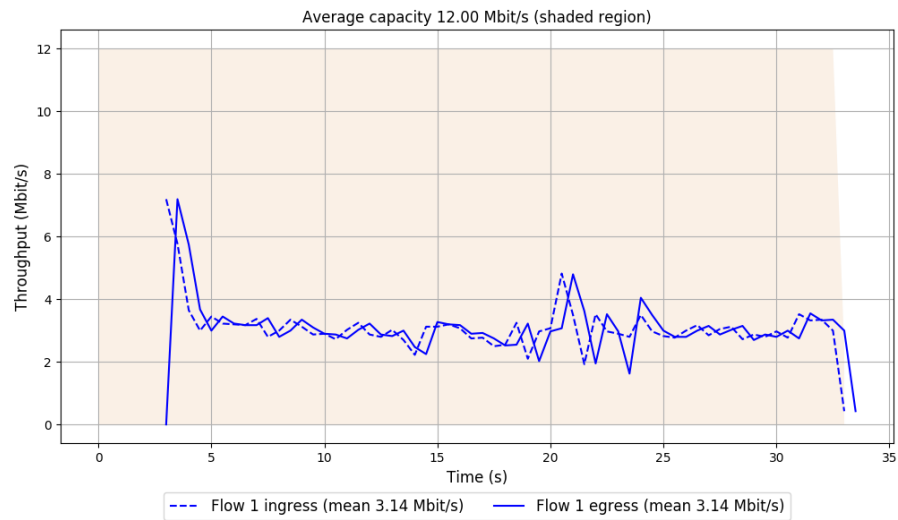
Figure is missing

```
Run 1: Statistics of TaoVA-100x

Start at: 2025-04-18 01:26:18
End at: 2025-04-18 01:26:48

# Below is generated by plot.py at 2025-04-18 01:32:06
# Datalink statistics
-- Total of 1 flow:
Average capacity: 12.00 Mbit/s
Average throughput: 3.14 Mbit/s (26.2% utilization)
95th percentile per-packet one-way delay: 61.102 ms
Loss rate: 0.00%
-- Flow 1:
Average throughput: 3.14 Mbit/s
95th percentile per-packet one-way delay: 61.102 ms
Loss rate: 0.00%
```

Run 1: Report of TaoVA-100x — Data Link



Run 1: Statistics of TCP Vegas

Start at: 2025-04-18 01:25:12

End at: 2025-04-18 01:25:42

Run 1: Report of TCP Vegas — Data Link

Figure is missing

Figure is missing

Run 1: Statistics of Verus

Start at: 2025-04-18 01:22:58

End at: 2025-04-18 01:23:28

Run 1: Report of Verus — Data Link

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Run 1: Statistics of PCC-Vivace

Start at: 2025-04-18 01:24:38

End at: 2025-04-18 01:25:08

Below is generated by plot.py at 2025-04-18 01:32:07

Datalink statistics

-- Total of 1 flow:

Average capacity: 12.00 Mbit/s

Average throughput: 5.71 Mbit/s (47.6% utilization)

95th percentile per-packet one-way delay: 35.577 ms

Loss rate: 0.01%

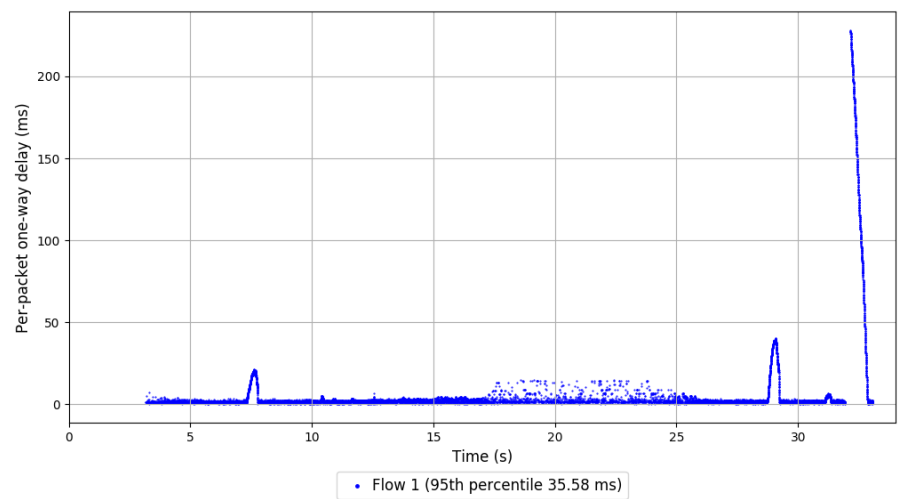
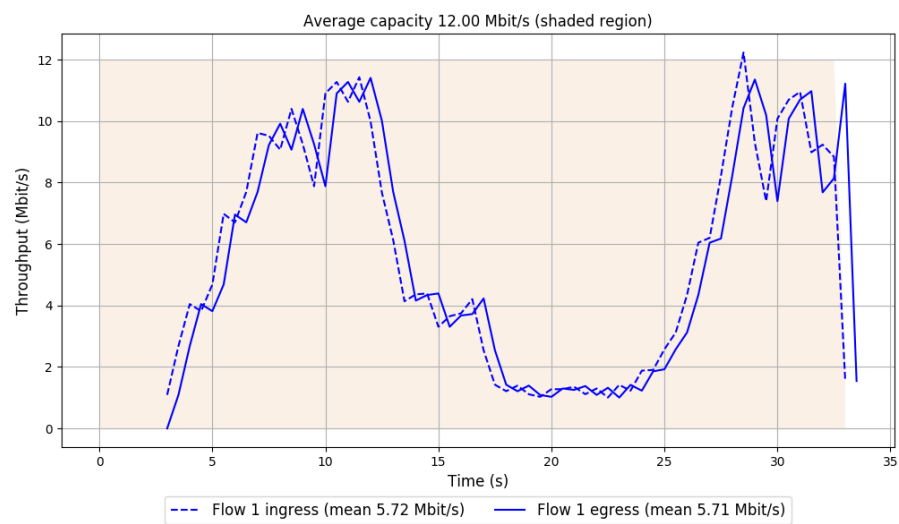
-- Flow 1:

Average throughput: 5.71 Mbit/s

95th percentile per-packet one-way delay: 35.577 ms

Loss rate: 0.01%

Run 1: Report of PCC-Vivace — Data Link



Run 1: Statistics of WebRTC media

Start at: 2025-04-18 01:29:06

End at: 2025-04-18 01:29:36

Run 1: Report of WebRTC media — Data Link

